GTC4Lusso USA - Checking and recharging battery - Latest Update: 2021-01-26

A3.26 Checking and recharging battery

Checking battery state of charge

Use a digital multimeter to measure the battery state of charge.

- Clean the battery poles and clamps adequately.
- Connect the terminals of the digital multimeter to the battery poles.
- Read the battery voltage value from the display of the digital multimeter.

If the battery voltage measured is **greater than 12.70 V**, the state of charge is **100%**.

If the battery voltage measured is **between 12.45 V and 12.70 V**, the state of charge is **75% to 100%**.

If the battery voltage measured is **between 12.36 V and 12.45 V**, the state of charge is **50% to 75%**.

If the battery voltage measured is **less than 12.36 V**, the state of charge is **0% to 50%**.

- Recharge the battery (A3.26).

Perform only if the battery voltage measured is **less than 12.5 V**.

Recharging the battery

Three charging methods are possible for "AGM" (gas recombination) batteries: FAST, SLOW and DEEP charging.

FAST charging is used for a partially or completely discharged battery

SLOW charging is used for a partially or completely discharged battery

DEEP charging is used for batteries that have discharged as a result of prolonged inactivity (over 2 weeks).

- Clean the battery poles and clamps adequately.
- Connect a stabilised battery charger to the battery.

FAST charging

While the battery is charging, check that the internal battery temperature never exceeds 45°C.
If the temperature exceeds 45°C, interrupt charging and wait a few hours before completing.

At the end of the recharge cycle, battery voltage must be at least 14.4 V - 14.8 V.
If this voltage is NOT reached (even if the battery was charged at constant current), this indicates a general deterioration in the state of the battery. In this case, replacing the battery is recommended even if the vehicle will start.

- Charge at a constant voltage of **14.4 V to 14.8 V**, at ambient temperature and with **limited initial charge current**.

For **L3 12V 70Ah 760A EN** batteries, the maximum initial charge current must be between **18 A** and **30 A**.

Charge cycle duration: **min. 4 hours**, **max. 6 hours**.

SLOW charging

While the battery is charging, check that the internal battery temperature never exceeds 45°C.
If the temperature exceeds 45°C, interrupt charging and wait a few hours before completing.

At the end of the recharge cycle, battery voltage must be at least 14.4 V - 14.8 V.
If this voltage is NOT reached (even if the battery was charged at constant current), this indicates a general deterioration in the state of the battery. In this case, replacing the battery is recommended even if the vehicle will start.

- Charge at a constant voltage of **14.4 V to 14.8 V**, at ambient temperature and with **limited initial charge current**.

For **L3 12V 70Ah 760A EN** batteries, the maximum initial charge current must be between **7 A** and **14 A**.

Charge cycle duration: **min. 12 hours**, **max. 24 hours**.

DEEP charging

While the battery is charging, check that the internal battery temperature never exceeds 45°C.
If the temperature exceeds 45°C, interrupt charging and wait a few hours before completing.

At the end of the recharge cycle, battery voltage must be at least 14.4 V - 14.8 V.
If this voltage is NOT reached (even if the battery was charged at constant current), this indicates a general deterioration in the state of the battery. In this case, replacing the battery is recommended even if the vehicle will start.

Prolonged periods with the battery not in use (more than 2 weeks with the battery connected) may cause faults at the next recharge cycle.
If the battery is charged in these high internal resistance conditions, the voltage may quickly reach the value of 14.4 - 14.8 V specified previously, but the current in the battery may drop rapidly and the battery may fail to maintain charge for the correct period.

- If the battery voltage is below 9 - 10 V, charge without **voltage limiting**.

The maximum initial charge current must be between **1 A** and **2 A**.

Charge cycle duration: **min. 12 hours**, **max. 24 hours**.

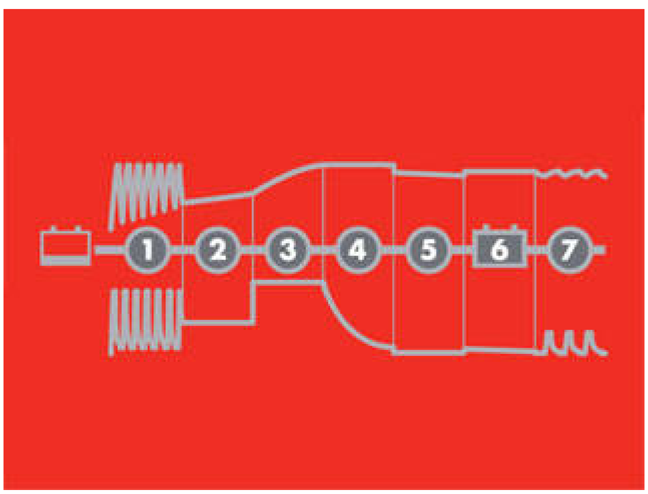
Battery conditioner

Ferrari vehicles are equipped with a battery conditioner with switched, pulse maintenance, buffer charging and analysis modes, representing the state of the art in battery care.
Ferrari battery conditioners are suitable for charging all types of 12V lead acid batteries: including liquid acid, MF, AGM, VRLA and the majority of gel batteries.

Technical specifications

Model: 1049
AC voltage: 220–240 V AC, 50–60Hz
Current absorption*: < 1mA
Nominal charging voltage: 12V
Fluctuation**: < 4%
Charging current: max 5A
Ambient temperature: -20 to +50°C, with automatic current reduction at higher temperatures.
Cooling: natural convection.
Charging cycle: Ferrari battery conditioners use a completely automatic charging cycle consisting of multiple stages
Battery types: all 12V lead acid batteries (liquid, Ca/Ca, MF, AGM and gel).
Battery capacity: from 14–160Ah
Dimensions: 168x65x38mm (L x W x H)
Insulation rating: IP65
Weight: 0.5 Kg
*) Return current absorption refers to the residual current in the battery when the battery conditioner is connected to the battery without connecting the mains power cable.
**) The stability of the current and voltage delivered is very important. High fluctuation values will result in battery overheating and cause accelerated deterioration of the positive electrode. High fluctuation values may also be harmful to the devices connected to the battery. The Ferrari battery conditioner delivers stable current with a very low degree of fluctuation.

Charging process stages



Ferrari battery conditioners use a completely automatic charging cycle consisting of multiple stages
Stage 1 - DESULPHATION: The battery conditioner determines if the battery is sulphated and, if necessary, delivers specific voltage and current pulses to eliminate the sulphates deposited on the lead plates of the battery and restore battery capacity.
Stage 2 - SOFT START: The battery conditioner determines if the battery is capable of being charged, to prevent attempting to charge a faulty battery.
Stage 3 - BULK: The battery conditioner charges with a current up to a maximum of 80% of battery capacity.
Stage 4 - ABSORPTION: The battery conditioner charges with decreasing current starting from 100% of battery capacity.
Stage 5 - ANALYSE: The battery conditioner tests if the battery is capable of maintaining charge. If not, it may be necessary to replace the battery.
Stage 6 - FLOAT: The battery conditioner maintains maximum battery voltage with a constant charging voltage.

Phase 7 - PULSE: The battery conditioner keeps the battery at 95–100% of capacity by monitoring the battery voltage and delivering a charge pulse when necessary to keep the battery fully charged.

Charging

- Connect the magnetic connector of the battery conditioner to the indicated socket on the luggage compartment lid.
- Connect the battery conditioner plug to the mains socket (the power lamp illuminates).
- Follow the progress of the 7 stages indicated on the display during the charging cycle.

The error warning lamp lights if the battery terminals are incorrectly connected.
The polarity inversion protection function prevents damage to the battery or battery conditioner in the event of incorrect connection.
When stage 4 is indicated, the battery is ready to start the engine.
When stage 6 is indicated, the battery is fully charged.
The charging cycle may be interrupted at any time by disconnecting the power cable from the mains socket.

Error modes

The battery conditioner enters error mode in the following conditions:

- Inverted polarity
- If the analyse function of the battery conditioner interrupts the charging process.
- If the terminals of the battery conditioner short-circuit when the charging cycle starts.
- If the battery conditioner remains in Start mode for more than 4 hours.

Press the RESET button to acknowledge/reset error mode.

Safety

The battery conditioner is designed to charge lead acid batteries with capacities from 14–160Ah. Do not use for any other purpose.
Ensure that the cables are not pinched and do not come into contact with hot surfaces or sharp edges.
The battery may release explosive gases during the charging cycle: do not allow sparks or naked flame near the battery.
The area must be adequately ventilated during the charging cycle.

Do not cover the battery conditioner.
Ensure that the mains socket is not exposed to water.
Never recharge a frozen battery.
Never recharge a damaged battery.
Never place the battery conditioner on the battery during the charging cycle.
Connections to mains electricity must comply with national law concerning high voltages.
Check the cables of the battery conditioner before use. Ensure that there are no cracks or splits in the cable insulation. Never use a battery conditioner with damaged cables.
Always ensure that the battery conditioner enters maintenance charging mode correctly before leaving unattended and in operation for prolonged periods of time. If the battery conditioner does not switch to maintenance charging mode within 72 hours, this means that a fault has occurred. Should this occur, disconnect the battery conditioner manually.
Eventually, all batteries wear out. If a battery fault is detected during charging, the battery is reconditioned by the advanced functions of the battery conditioner. Some faults, however, may not be correctable by the battery conditioner. Never leave the battery charger unattended and operating for prolonged periods of time.
Place on a flat surface only.
This device must not be operated by children or persons who cannot read and understand the indications of this manual unless supervised by an adult ensuring that the battery conditioner is used safely. Store and use the battery conditioner out of the reach of children; never let children play with the battery conditioner.

Overheating protection

The Ferrari battery conditioner has an overheat protection function.
Output power is reduced in the event of increased ambient temperature.

Maintenance

The Ferrari battery conditioner requires no maintenance.
Never dismantle the battery conditioner. Any dismantling or tampering will void the warranty.
Take the battery charger to the dealer for repair if the power cable is damaged.
The cover of the battery conditioner may be cleaned with a soft, dampened cloth and neutral detergent.
Always disconnect the Ferrari battery conditioner from the mains before cleaning.